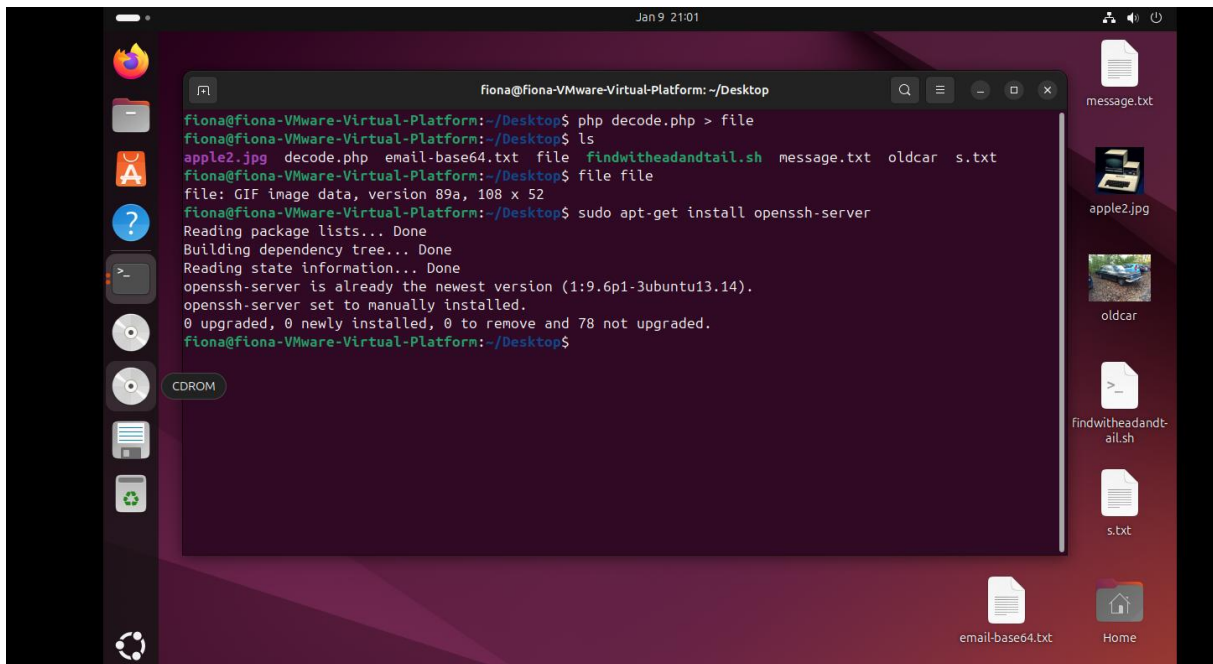


Template Week 6 – Networking

Student number: 583809

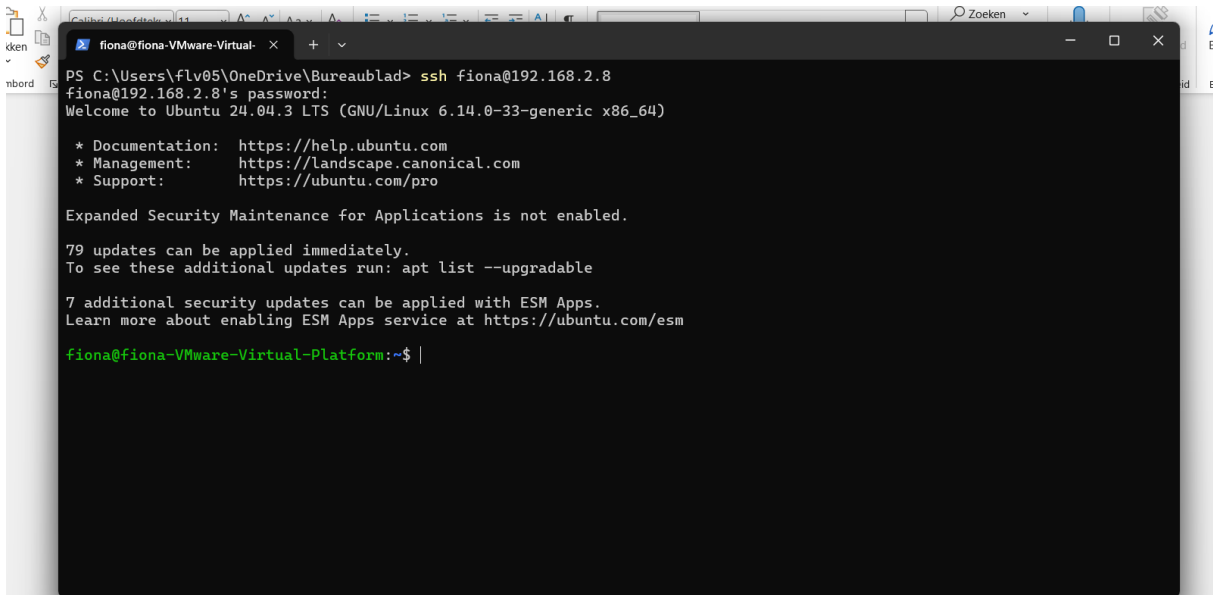
Assignment 6.1: Working from home

Screenshot installation openssh-server:



```
fiona@fiona-VMware-Virtual-Platform: ~/Desktop
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ php decode.php > file
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ ls
apple2.jpg  decode.php  email-base64.txt  file  findwithheadandtail.sh  message.txt  oldcar  s.txt
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ file file
file: GIF image data, version 89a, 108 x 52
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ sudo apt-get install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:9.6p1-3ubuntu13.14).
openssh-server set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 78 not upgraded.
fiona@fiona-VMware-Virtual-Platform:~/Desktop$
```

Screenshot successful SSH command execution:



```
PS C:\Users\flv05\OneDrive\Bureaublad> ssh fiona@192.168.2.8
fiona@192.168.2.8's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-33-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

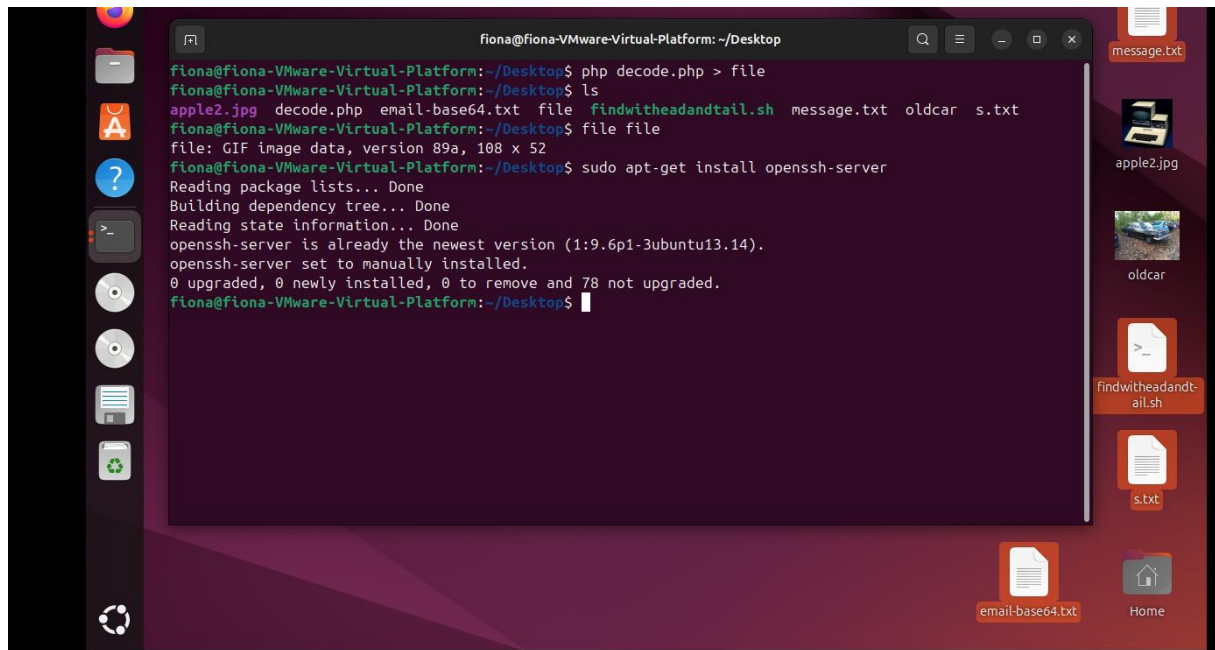
79 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

7 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

fiona@fiona-VMware-Virtual-Platform:~$
```

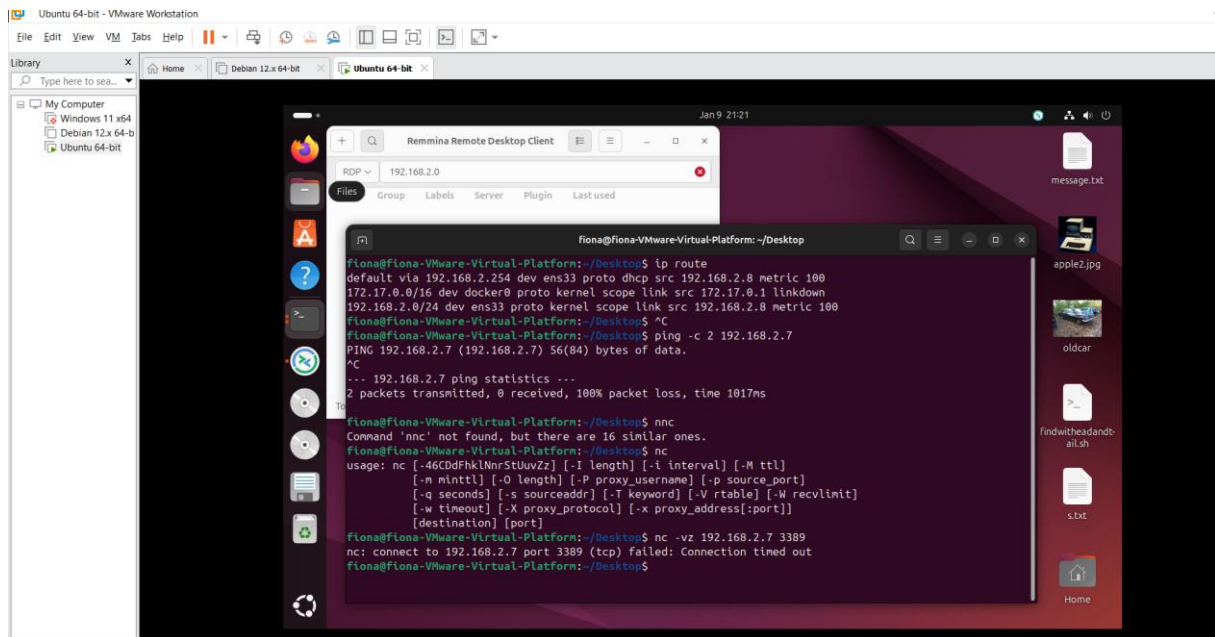
Screenshot successful execution SCP command:

```
Connection to 192.168.2.8 closed.
PS C:\Users\flv05\OneDrive\Bureaublad> scp .\email-base64.txt fiona@192.168.2.8:~/Desktop
fiona@192.168.2.8's password:
email-base64.txt                                     100% 2509   490.0KB/s   00:00
PS C:\Users\flv05\OneDrive\Bureaublad> |
```



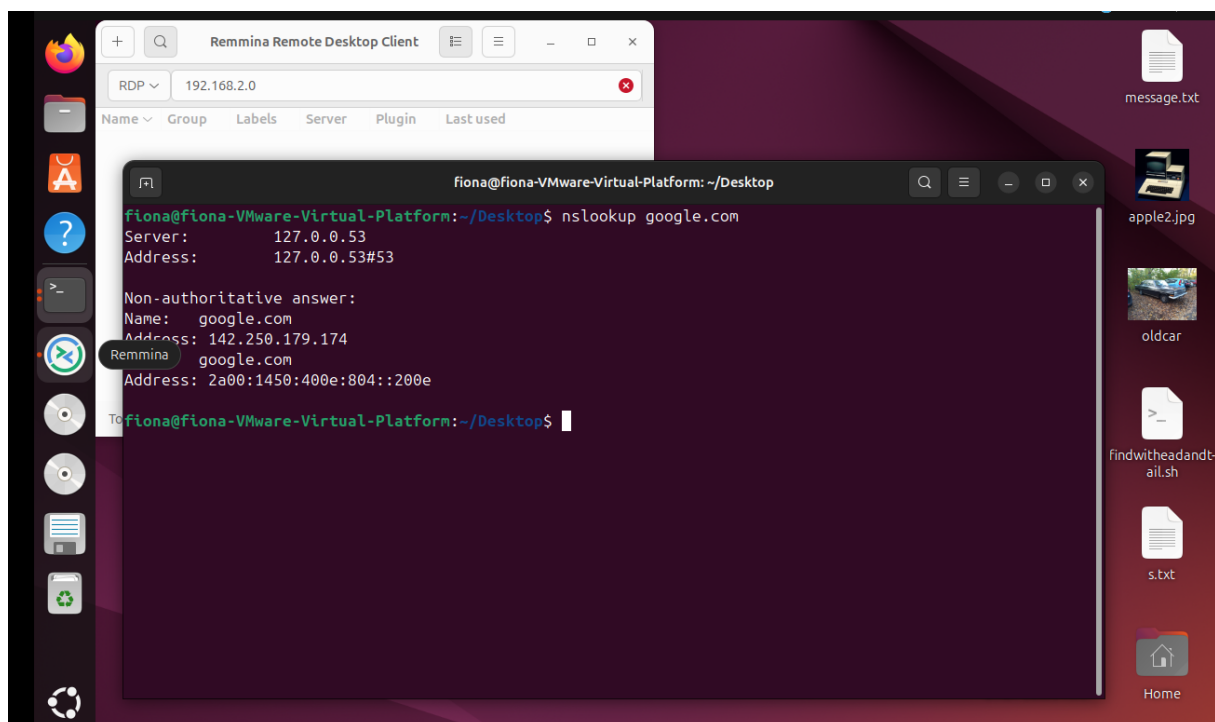
Screenshot remmina:

```
-----update package remmina
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ sudo apt-get install remmina
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libavahi-ui-gtk3-0 libfreerdp-client3-3 libvncclient1 remmina-common remmina-plugin-rdp
  remmina-plugin-secret remmina-plugin-vnc
Suggested packages:
  remmina-plugin-exec remmina-plugin-kwallet remmina-plugin-python remmina-plugin-spice
  remmina-plugin-www remmina-plugin-x2go
The following NEW packages will be installed:
```



Assignment 6.2: IP addresses websites

Relevant screenshots nslookup command:



```
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ nslookup amazon.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   amazon.com
Address: 98.82.161.185
Name:   amazon.com
Address: 98.87.170.74
Name:   amazon.com
Address: 98.87.170.71
```

```
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ nslookup one.one.one.one
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   one.one.one.one
Address: 1.0.0.1
Name:   one.one.one.one
Address: 1.1.1.1
Name:   one.one.one.one
Address: 2606:4700:4700::1111
Name:   one.one.one.one
Address: 2606:4700:4700::1001
```

```
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ nslookup dns.google.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   dns.google.com
Address: 8.8.8.8
Name:   dns.google.com
Address: 8.8.4.4
Name:   dns.google.com
Address: 2001:4860:4860::8844
Name:   dns.google.com
Address: 2001:4860:4860::8888
```

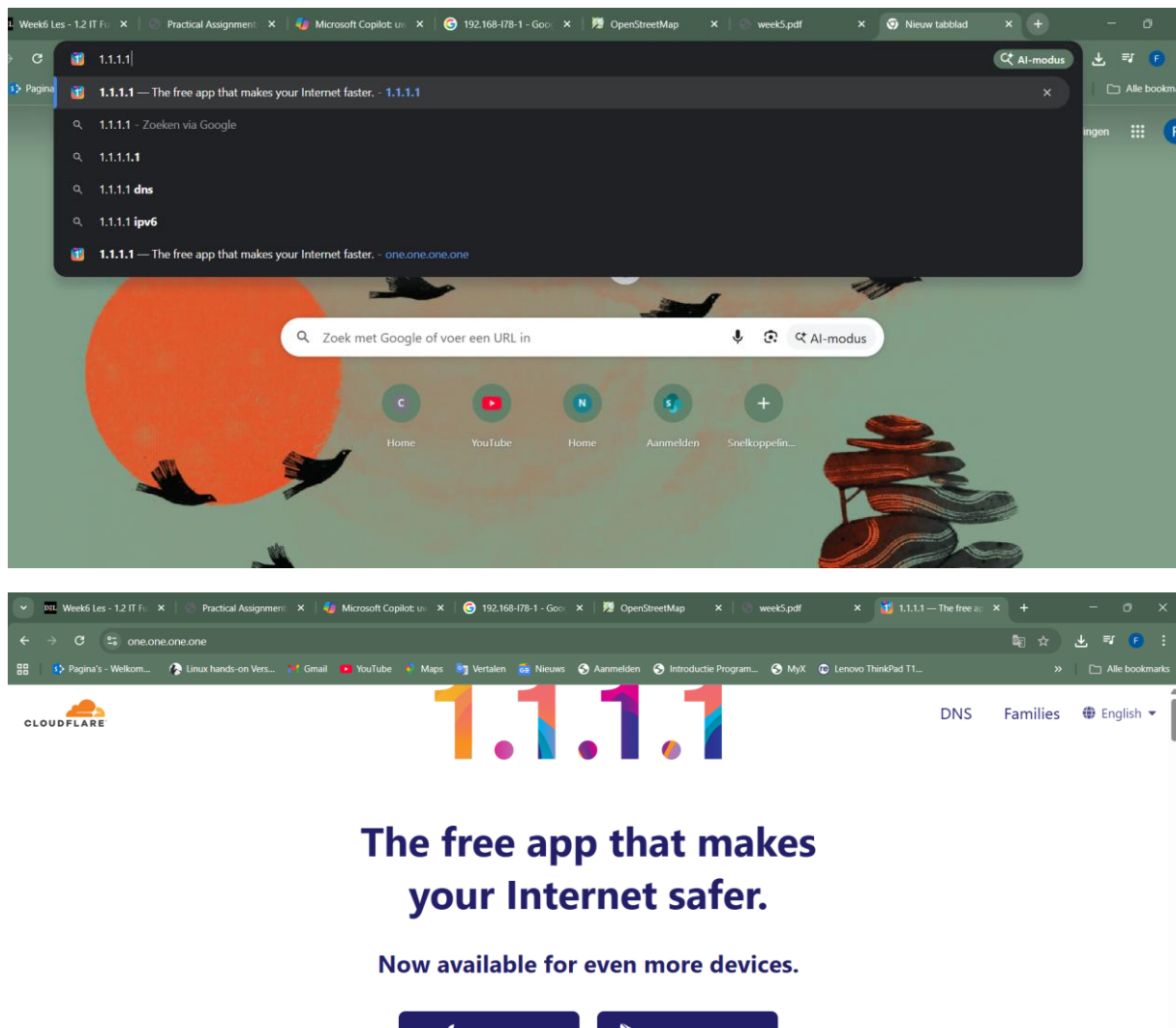
```
fiona@fiona-VMware-Virtual-Platform:~/Desktop$ nslookup bol.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   bol.com
Address: 79.170.100.62
```

```
ROM
@fiona-VMware-Virtual-Platform:~/Desktop$ nslookup w3schools.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   w3schools.com
Address: 13.248.240.135
Name:   w3schools.com
Address: 76.223.115.82
```

Screenshot website visit via IP address:



Assignment 6.3: subnetting

How many IP addresses are in this network configuration 192.168.110.128/25?

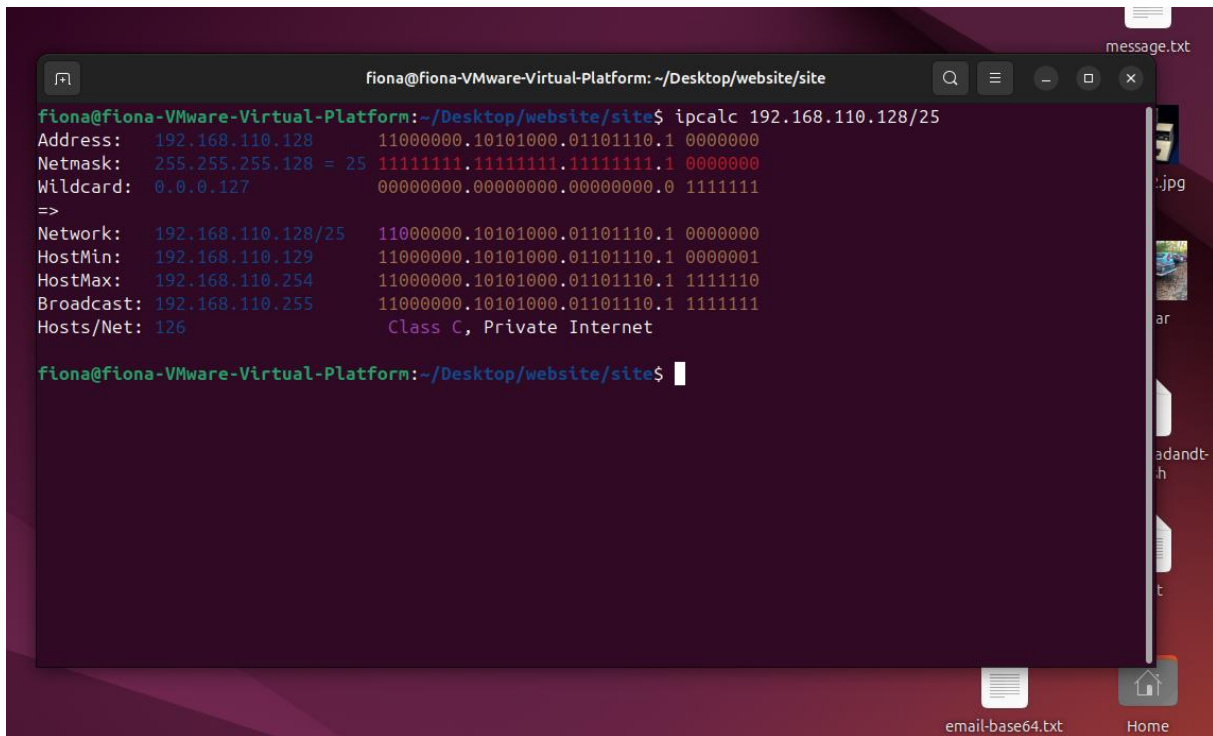
128 adressen dat komt doordat er 32 bits in een ip adres zit. 25 bits voor het netwerk gedeelte en 7 voor het host gedeelte. 2^7 is dan 128.

Zoals we weten kan elke bit 0 of 1 zijn dus dat wordt dan 2^7 dat zijn dan 128 adressen. Maar er zijn twee adressen onbruikbaar

What is the usable IP range to hand out to the connected computers?

192.168.110.129 → 192.168.110.254

Check your two previous answers with this Linux command: `ipcalc 192.168.110.128/25`



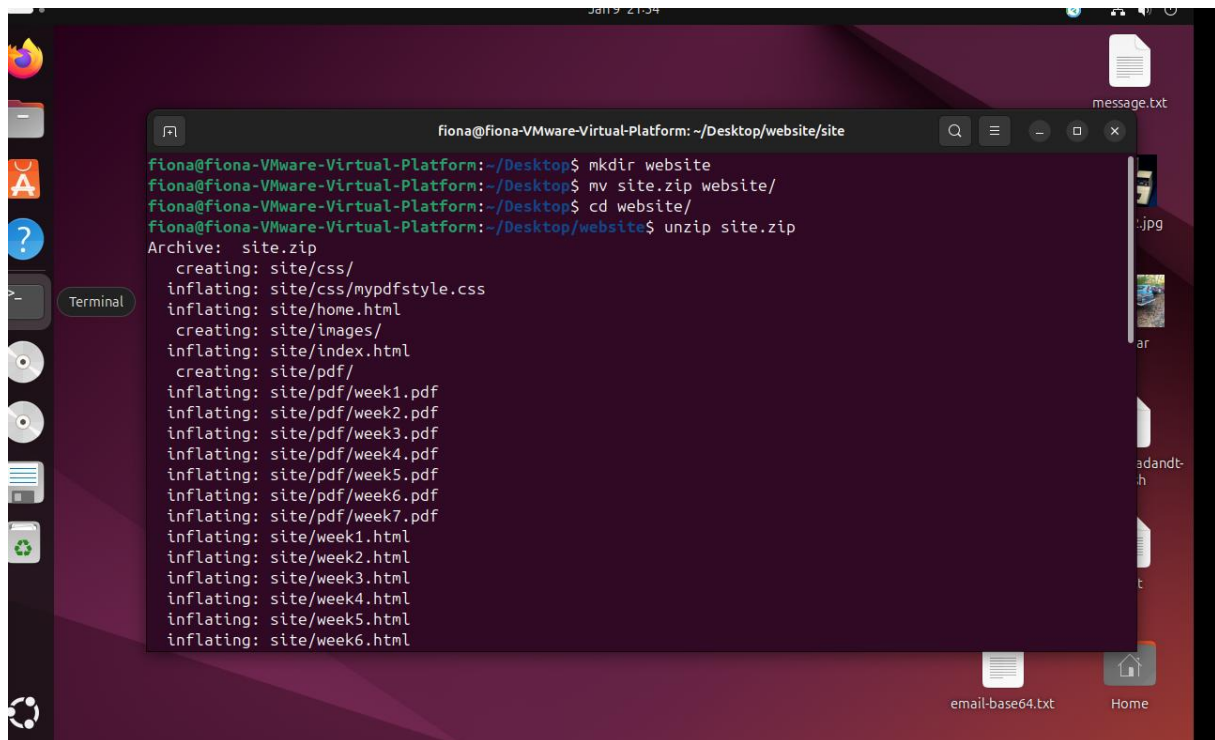
The screenshot shows a terminal window titled "fiona@fiona-VMware-Virtual-Platform: ~/Desktop/website/site". The command `ipcalc 192.168.110.128/25` has been executed, resulting in the following output:

```
fiona@fiona-VMware-Virtual-Platform:~/Desktop/website/site$ ipcalc 192.168.110.128/25
Address: 192.168.110.128      11000000.10101000.01101110.1 0000000
Netmask: 255.255.255.128 = 25 11111111.11111111.11111111.1 0000000
Wildcard: 0.0.0.127          00000000.00000000.00000000.0 1111111
=>
Network: 192.168.110.128/25  11000000.10101000.01101110.1 0000000
HostMin: 192.168.110.129    11000000.10101000.01101110.1 0000001
HostMax: 192.168.110.254    11000000.10101000.01101110.1 1111110
Broadcast: 192.168.110.255  11000000.10101000.01101110.1 1111111
Hosts/Net: 126              Class C, Private Internet

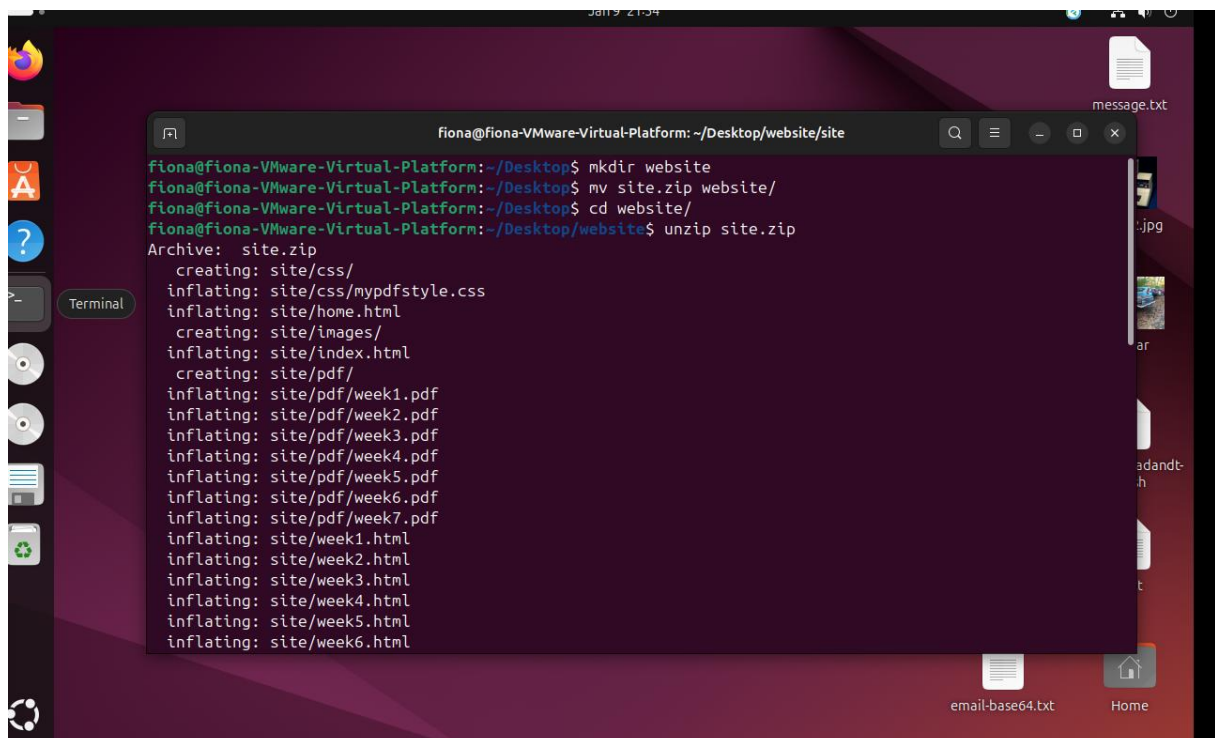
fiona@fiona-VMware-Virtual-Platform:~/Desktop/website/site$
```

Assignment 6.4: HTML

Screenshot IP address Ubuntu VM:



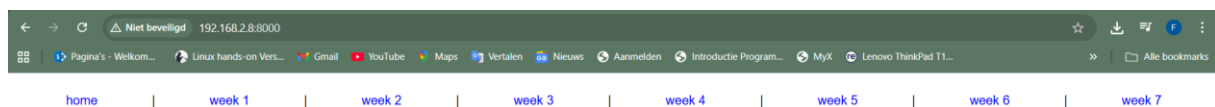
Screenshot of Site directory contents:



Screenshot python3 webserver command:

```
Fiona@fiona-VMware-Virtual-Platform: ~/Desktop/website/site
inflating: site/week4.html
inflating: site/week5.html
inflating: site/week6.html
inflating: site/week7.html
fiona@fiona-VMware-Virtual-Platform:~/Desktop/website$ ls
site  site.zip
fiona@fiona-VMware-Virtual-Platform:~/Desktop/website$ rm site.zip
fiona@fiona-VMware-Virtual-Platform:~/Desktop/website$ cd site
fiona@fiona-VMware-Virtual-Platform:~/Desktop/website/site$ ls
css      images      pdf          week2.html  week4.html  week6.html
home.html index.html  week1.html  week3.html  week5.html  week7.html
fiona@fiona-VMware-Virtual-Platform:~/Desktop/website/site$ python
python3      pythoncalls-bpfcc  pythongc-bpfcc
python3.12   pythonflow-bpfcc   pythonstat-bpfcc
fiona@fiona-VMware-Virtual-Platform:~/Desktop/website/site$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
192.168.2.7 - - [09/Jan/2026 21:54:16] code 400, message Bad request version ('"4ZË;óç\\x1a$YU\\x01\\x86
\\x82\\x81N\\x9d\\x03\\x91â\\x9b_0g`i')
192.168.2.7 - - [09/Jan/2026 21:54:16] "\x16\x03\x01\x07\x00\x01\x00\x06ü\x03\x03d7,\x9b_ \x92q\x14öeÿi\
x17æA\x9dô\x91(I\x9a-\x1e\x96\x84|ñj`$İ üiu\x85\x0bûn,p>ç\x11ÊTJ\x83\x8eæj²\x93L_ \x09\x87#2\x17`ñ\x00
ÚÚ\x13\x01\x13\x02\x13\x03Â+Â/Â,Â0İeİ`Â\x13Â\x14\x00\x9c\x00\x9d\x00/\x005\x01\x00\x06\x93\x9a\x9a\x00\x
00\x00-\x00\x02\x01\x01\x00\x17\x00\x00\x00#\x00\x00\x00\x1b\x00\x03\x02\x00\x02DÍ\x00\x05\x00\x03\x02h2
b\x0d\x01\x1a\x00\x00\x01\x00\x01*\x00 äy\x8aU\x95±¼Çxat\x84\x11\x80xuÂİÇíÂ\x8aÊôP\x0330|\x8di\x1e\x00ô`
Â\x86ô,zé,BäC$z}\x03Kc+=\x06\x0cê` \x90\x94\x00\x90ë\x13N²rC \x86\x95CÔ¶4H@İM9iQ+\x82itôP\x05\x89ü` Ö|EB\
```

Screenshot web browser visits your site



Hello World!

Assignment 6.5: Network segment

Remember that bitwise java application you've made in week 2? Expand that application so that you can also calculate a network segment as explained in the PowerPoint slides of week 6. Use the bitwise & AND operator. You need to be able to input two Strings. An IP address and a subnet.

IP: 192.168.1.100 and subnet: 255.255.255.224 for /27

Example: 192.168.1.100/27

Calculate the network segment

IP Address: 11000000.10101000.00000001.01100100

Subnet Mask: 11111111.11111111.11111111.11100000

Network Addr: 11000000.10101000.00000001.01100000

This gives 192.168.1.96 in decimal as the network address.

For a /27 subnet, each segment (or subnet) has 32 IP addresses (2^5).

The range of this network segment is from 192.168.1.96 to 192.168.1.127.

Paste source code here, with a screenshot of a working application.



```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
  
        System.out.println("--- Bitwise Programma ---");  
        System.out.println("1. Oneven check");  
        System.out.println("2. Macht van 2 check");  
        System.out.println("3. Two's complement");  
        System.out.println("4. Bereken netwerksegment");  
        System.out.print("Maak een keuze: ");
```

```
        int choice = scanner.nextInt();  
        scanner.nextLine(); // buffer leegmaken
```

```
        switch (choice) {
```

```

case 1:
    System.out.print("Voer een getal in: ");
    int number = scanner.nextInt();
    if ((number & 1) == 1)
        System.out.println(number + " is oneven");
    else
        System.out.println(number + " is even");
    break;

case 2:
    System.out.print("Voer een getal in: ");
    int num = scanner.nextInt();
    if (num > 0 && (num & (num - 1)) == 0)
        System.out.println(num + " is een macht van 2");
    else
        System.out.println(num + " is GEEN macht van 2");
    break;

case 3:
    System.out.print("Voer een getal in: ");
    int n = scanner.nextInt();
    int twosComp = ~n + 1;
    System.out.println("Two's complement van " + n + " = " + twosComp);
    break;

case 4:
    System.out.print("Voer een IP-adres in (bijv. 192.168.1.100): ");
    String ip = scanner.nextLine();

    System.out.print("Voer een subnetmasker in (bijv. 255.255.255.224): ");
    String subnet = scanner.nextLine();

    int network = calculateNetwork(ip, subnet);
    System.out.println("Netwerkadres = " + intToIP(network));
    break;

default:
    System.out.println("Ongeldige keuze");
}

scanner.close();
}

// ---- Functies ----

// Zet IP string om naar 32-bit integer
public static int ipToInt(String ip) {
    String[] parts = ip.split("\\.");

```

```

        return (Integer.parseInt(parts[0]) << 24) |
            (Integer.parseInt(parts[1]) << 16) |
            (Integer.parseInt(parts[2]) << 8) |
            Integer.parseInt(parts[3]);
    }

    // Zet 32-bit integer terug naar IP string
    public static String intToIP(int value) {
        return ((value >> 24) & 0xFF) + "." +
            ((value >> 16) & 0xFF) + "." +
            ((value >> 8) & 0xFF) + "." +
            (value & 0xFF);
    }

    // Bereken netwerksegment met bitwise AND
    public static int calculateNetwork(String ip, String subnet) {
        int ipInt = ipToInt(ip);
        int maskInt = ipToInt(subnet);
        return ipInt & maskInt; // BITWISE AND
    }
}

```

Ready? Save this file and export it as a pdf file with the name: [week6.pdf](#)